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পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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SUB-SOIL INVESTIGATION REPORT

FOR THE PROPOSED
Six Storied Residential Building

AT SYEDPUR, MEJORTILA, SYLHET.
DAG # 290,291, KHA # 8602, J.L # 96, MOUJA- DEVPUR

CLIENT

JAHANARA BEGUM
W/O- MD. ENAMUL HOQUE

Verification Code: NWep33w0
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTED BY

M/S RELATION SOIL INVESTIGATION
Kumarpara, Sylhet.
Mob : 01711063018 & 01712186249

Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would be necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-30 ft	06
02	Disturbed sample	02	0-25 ft	05
03	Disturbed sample	03	0-25 ft	05
04	Disturbed sample	04	0-25 ft	05
05	Disturbed sample	05	0-25 ft	05
06	Disturbed sample	06	0-35 ft	07
07	Disturbed sample	07	0-25 ft	05

Laboratory Classifications

The disturbed soil samples supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N;$ $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α – Adhesion factor, f_{su} – Ultimate skin friction, C_u – Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = 4N/200\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)**

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	25	2.946	3.099	3.378	3.480	3.759	4.089	3.175
	5.0-10	39	4.472	4.703	5.127	5.282	5.706	6.207	4.819
BH02	2.5-5.0	47	5.429	5.710	6.224	6.412	6.926	7.535	5.850
	5.0-10	56	6.516	6.853	7.471	7.696	8.314	9.044	7.022
BH03	2.5-5.0	41	4.802	5.051	5.506	5.672	6.127	6.665	5.175
	5.0-10	52	6.044	6.357	6.930	7.138	7.711	8.389	6.513
BH04	2.5-5.0	41	4.779	5.026	5.480	5.644	6.098	6.633	5.150
	5.0-10	51	5.920	6.226	6.788	6.992	7.553	8.217	6.379
BH05	2.5-5.0	38	4.454	4.685	5.107	5.261	5.683	6.182	4.800
	5.0-10	49	5.700	5.995	6.535	6.732	7.272	7.911	6.142
BH06	2.5-5.0	22	2.598	2.733	2.979	3.069	3.315	3.606	2.800
	5.0-10	35	4.116	4.329	4.719	4.861	5.251	5.713	4.435
BH07	2.5-5.0	45	5.174	5.441	5.932	6.110	6.601	7.181	5.575
	5.0-10	54	6.240	6.563	7.154	7.370	7.961	8.661	6.724
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

** Designer should consider both the calculated design N value and settlement behavior of the corresponding soil profile while selecting shallow footings

Table 8: The allowable skin friction for precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.074	0.241	0.156	0.096	0.049	0.049	0.240	-	-	-
10	0.261	0.682	0.655	0.669	0.602	0.281	0.669	-	-	-
15	0.669	0.762	0.682	0.682	0.669	0.455	0.709	-	-	-
20	0.682	0.803	0.749	0.696	0.696	0.669	0.776	-	-	-
25	0.749	0.843	0.816	0.829	0.762	0.294	0.883	-	-	-
30	0.776	-	-	-	-	0.736	-	-	-	-
35	-	-	-	-	-	0.856	-	-	-	-
40	-	-	-	-	-	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for precast driven piles in Tsf (FS=3.0)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.675	4.050	1.575	0.900	0.450	0.450	3.825	-	-	-
10	2.700	11.475	11.025	11.250	10.125	4.725	11.250	-	-	-
15	11.250	12.825	11.475	11.475	11.250	7.650	11.925	-	-	-
20	11.475	13.500	12.600	11.700	11.700	11.250	13.050	-	-	-
25	12.600	14.175	13.725	13.950	12.825	4.950	14.850	-	-	-
30	13.050	-	-	-	-	12.375	-	-	-	-
35	-	-	-	-	-	14.400	-	-	-	-
40	-	-	-	-	-	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.048	0.156	0.101	0.062	0.032	0.032	0.156	-	-	-
10	0.170	0.443	0.426	0.435	0.391	0.183	0.435	-	-	-
15	0.435	0.496	0.443	0.443	0.435	0.296	0.461	-	-	-
20	0.443	0.522	0.487	0.452	0.452	0.435	0.504	-	-	-
25	0.487	0.548	0.530	0.539	0.496	0.191	0.574	-	-	-
30	0.504	-	-	-	-	0.478	-	-	-	-
35	-	-	-	-	-	0.556	-	-	-	-
40	-	-	-	-	-	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.304	1.823	0.709	0.405	0.203	0.203	1.721	-	-	-
10	1.215	5.164	4.961	5.063	4.556	2.126	5.063	-	-	-
15	5.063	5.771	5.164	5.164	5.063	3.443	5.366	-	-	-
20	5.164	6.075	5.670	5.265	5.265	5.063	5.873	-	-	-
25	5.670	6.379	6.176	6.278	5.771	2.228	6.683	-	-	-
30	5.873	-	-	-	-	5.569	-	-	-	-
35	-	-	-	-	-	6.480	-	-	-	-
40	-	-	-	-	-	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the construction of the proposed six storied residential building.

Soil profile scenario

Overall soil information of the investigated site is quite regular in between the boreholes. However, vertical inconsistency exists in the prevailing soil formation. In the boreholes 1, 3, 4, 5 and 6 clay soil of soft to very soft consistency has been observed up to 10 ft depth followed by stiff to hard clay layer up to 25-35 ft depth. However, in the boreholes 2 and 7, stiff to hard clay layer has been observed up to 25-35 ft depth.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed boundary wall. Since the whole site is dominated by cohesive soil with soft to very soft consistency in the top 10 ft depth, shallow foundation without ground improvement is not suitable for this type of structure. In this regard, the whole site in accordance with the borehole condition may be densified by sand replacement method associated with proper compaction. The depth of ground improvement should be selected by the foundation/design engineer considering both pressure influence zone and degree of densification required to minimize uneven settlement. After soil improvement, appropriate pattern of shallow foundation may be selected on the basis of degree of ground improvement. Although, bearing capacity values for designing shallow foundation can be found from the attached table (7) in the report which is based on existing soil condition; it is necessary to reinvestigate the modified bearing capacity of the densified stratum after ground improvement.

Safety provisions

Since al the boreholes of the proposed site are dominated with cohesive soil, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the settlement of the site considering building load while designing either shallow or deep foundation. Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. No further investigation was done by the client.

Special notes

These suggestions are absolutely tentative, and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **25-35 ft** as client provided.

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